

Training Novices to Provide Robot-Learnable Visuomotor Demonstrations

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Abstract: This paper introduces deployment-evaluated machine teaching (DEMT), a pre-collection training framework for improving novice demonstrations in visuomotor imitation learning. Scalable teleoperation and augmented reality (AR) interfaces ease data collection, but task-successful demonstrations may still provide poor supervision for a fixed learner under a finite budget. DEMT defines demonstration quality by downstream robot deployment and trains spatial, contact, and temporal consistency through scaffolded AR guidance. Controlled simulations show that inconsistent successful demonstrations reduce deployment success by an average of 70 percentage points relative to the DEMT baseline. In a study involving teaching a Franka Panda robot, DEMT increases deployment success from 14% to 75% on the trained task and from 22% to 91% on an unseen object. On an unseen longer-horizon ordered pick-and-place task, DEMT-trained users achieve a 55-percentage-point higher deployment success rate than the control group.

Keywords: Visuomotor Policy, Human Training, Augmented Reality

1 Introduction

Recent imitation learning or learning from demonstration (LfD) methods have enabled strong progress in visuomotor robot manipulation, for example, through Diffusion Policies and the Action Chunking Transformer [1, 2]. However, the effectiveness of imitation learning depends not only on the policy architecture, but also on the demonstrations, which need to be of sufficient quantity and quality. Regarding data quantity, recent in-the-wild and low-cost teleoperation systems have made large-scale robot demonstration collection cheaper, more accessible, and easier to scale [3, 4, 5, 6, 7, 8]. However, scalable collection addresses the cost of obtaining demonstrations, not whether those demonstrations are equally useful for robot learning.

As to demonstration quality, a common assumption is that a high-quality demonstration is a successful one. However, this assumption is insufficient because human demonstrators rely on sensing, dexterity, tactile feedback, semantic reasoning, and online correction that are *not available to the robot*. The robot must infer an observation-to-action mapping under its own embodiment, perception, controller, policy class, and finite data budget [9, 10, 11, 12, 13].

This paper focuses on the question of whether *novice users can be trained to provide high-quality demonstrations for efficient robot learning* in the face of this mismatch in human and robot capabilities. Specifically, it introduces *deployment-evaluated machine teaching (DEMT)*, a framework for selecting guidance and training novice users. DEMT identifies and scaffolds demonstrations to make behaviours more learnable for the robot. In the context of visuomotor manipulation, the DEMT framework defines spatio-contact-temporal guidance structures [14, 15, 10, 11] that provide more robot-learnable demonstrations, that enable users to better teach manipulation tasks. Experiments show that users trained through DEMT provide more robot-learnable demonstrations after guidance is removed, increasing deployment success from 14% to 75% on the trained task and from

22% to 91% on an unseen object. On a longer-horizon ordered pick-and-place task, policies trained from DMT-trained users achieve 55% deployment success, while the control group were unable to teach the task with any success.

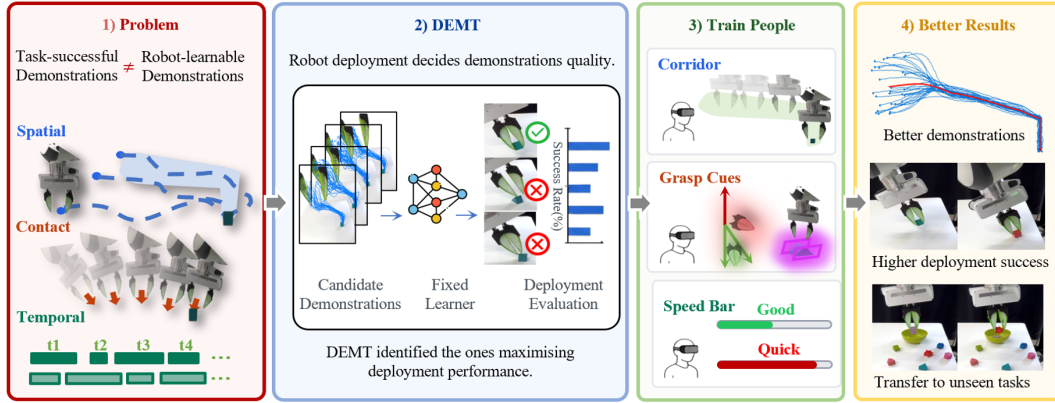


Figure 1: Overview of the DMT framework. Task-successful human demonstrations are not assumed to be equally learnable: spatial, contact, and temporal variations can still lead to poor robot learning. DMT evaluates demonstration quality through robot deployment under a fixed learner and finite budget, and identifies the learner-relevant structures that improve deployment performance. These structures are translated into human training guidance, for instance, corridor, grasp-cue, and speed-bar scaffolds. After guidance is removed, trained users provide better unguided demonstrations, leading to higher deployment success and transfer to unseen tasks.

Related Work

Scaling Human Demonstration Quantity

Recent robot data-collection systems make manipulation demonstrations easier to collect beyond controlled laboratories. UMI [3] supports portable in-the-wild collection with low-cost handheld hardware, while Mobile ALOHA [6] extends low-cost teleoperation to mobile bimanual manipulation. Open Teach [4], DexCap [5], and EgoMI [8] further lower the interface barrier through mixed-reality teleoperation, dexterous motion capture, and egocentric retargeting. These systems allow increased quantity of demonstrations. In contrast, the work presented here addresses a complementary problem: how novice demonstrators can be trained so that collected demonstrations are more learnable for the robot.

Improving Demonstration Quality

Related work studies how demonstration quality can be analysed or improved at the data level. Data-centric studies characterise demonstration quality through factors such as action divergence, transition diversity, consistency, and trajectory predictability [16, 10, 11, 12]. Post-hoc curation methods further improve datasets through filtering, reweighting, relabelling, demonstration selection, or mixture optimisation [17, 18, 19, 20, 21, 13]. These approaches improve demonstration quality at the data level, without explicitly changing the behaviour of the human demonstrator; this paper instead improves demonstration quality by improving human teaching.

Improving Human Teaching

Another line of work improves demonstrations by shaping human behaviour during and before collection. AR/XR systems such as ARCap [7], ARMimic [22], and ARMADA [23] provide real-time feedback on feasibility, visibility, and collisions, while related systems use feasibility cues, force feedback, or refinement to improve execution [24, 25, 26, 27]. These methods rely on feedback being active during collection.

This paper focuses on pre-collection human training: whether novice users retain better behaviour after guidance is removed. Building on machine teaching (MT) [28] and prior work on retained,

transferable teaching behaviour [9, 29, 30], this paper extends pre-collection training from low-dimensional proprioceptive skills to visuomotor manipulation, where quality is evaluated by robot deployment rather than human task success alone.

2 Problem Definition

This paper considers problems involving visuomotor policies, such as 3D Diffusion Policies [1, 5, 7]. Specifically, visuomotor tasks are communicated to the robot learner by teaching, *i.e.*, by providing a set of demonstrations $\mathcal{D}$, consisting of observations ($\mathbf{O} \in \mathbb{O}$) and actions ($\mathbf{A} \in \mathbb{A}$), from the space of possible data sets $\mathbb{D}$. The robot learner learns the visuomotor tasks with learning algorithm $\mathcal{A}(\cdot)$ leading to the learnt model

$$\tilde{\theta} = \mathcal{A}(\mathcal{D}), \quad \mathcal{D} = (\mathbf{O}, \mathbf{A}). \quad (1)$$

For efficient robot learning, improving LfD depends on the quality of the demonstrations provided to the learner. Under a finite demonstration budget, the problem therefore becomes identifying which demonstrations best induce the desired model θ^* . In MT[28], this is formulated as finding the optimal teaching set $\mathcal{D}^*$ through the following bi-level optimisation problem:

$$\mathcal{D}^* = \arg \min_{\mathcal{D} \in \mathbb{D}} \rho(\tilde{\theta}, \theta^*) \quad \text{s.t.} \quad \tilde{\theta} = \mathcal{A}(\mathcal{D}) \quad \text{and} \quad \varepsilon(\mathcal{D}) = N. \quad (2)$$

The teaching risk $\rho(\cdot)$ measures how well the model learned from $\mathcal{D}$ recovers the target behaviour represented by θ^* . The $\varepsilon(\cdot)$ represents the teaching budget. In this paper, N denotes the minimum demonstration budget, and MT seeks the teaching set that minimises the teaching risk $\rho(\cdot)$ under this budget. However, the closed-form solution of such bi-level optimisation requires *the target model θ^* to be explicitly known* in order to compute $\rho(\cdot)$. In practice, this condition is rarely met for modern visuomotor learning approaches, where the algorithm is represented by a high-dimensional neural architecture with visual encoders and action prediction networks.

3 Deployment-Evaluated Machine Teaching Framework

3.1 DMT Formulation

To address this, DMT is proposed to make teaching risk observable through deployment. Specifically, DMT replaces $\rho(\cdot)$ with a deployment performance objective J . The latter is written as

$$J(\mathcal{A}(\mathcal{D})) = \mathbb{E}[\mathbf{S}(\mathcal{A})] \quad (3)$$

where, $\mathbf{S}(\mathcal{A})$ denotes the learner deployment evaluation protocol. This is evaluated by deployment success together with general deployment constraints such as safe execution, stable contact, and absence of damaging collisions.

To make J in (3) actionable for human training, this work restricts the teaching search to human-controllable structures in visuomotor manipulation, informed by prior LfD analyses of demonstration quality [16, 10, 11, 12]. This gives the structured teaching space $\mathcal{Z} = \mathcal{P} \times \mathcal{R} \times \mathcal{V}$:

s_1 **Spatial structure** ($\mathcal{P}$): the geometric path followed by the end effector, specifying *where* the robot moves.

s_2 **Contact structure** ($\mathcal{R}$): the contact or grasp strategy used when interacting with the target object, specifying *how* the robot interacts.

s_3 **Temporal structure** ($\mathcal{V}$): the speed profile and phase timing around task-relevant events, specifying *when* critical actions occur.

Let $z = (p, r, v) \in \mathcal{Z}$ denote a concrete teaching configuration, which determines how demonstrations vary in spatial paths, contact strategies, event alignment, and phase-wise velocity profiles. Given a fixed learner, embodiment, sensing pipeline, and demonstration budget, DMT evaluates which teaching configuration yields the highest deployment performance:

$$z^* = \arg \max_{z \in \mathcal{Z}} J(\mathcal{A}(\mathcal{D}_z)) \quad \text{s.t.} \quad |\mathcal{D}_z| = N. \quad (4)$$

Here, $\mathcal{D}_z$ is the dataset induced by demonstrations collected under teaching configuration z and z^* denotes the deployment-selected teaching configuration within the $\mathcal{P}$ - $\mathcal{R}$ - $\mathcal{V}$ structure space.

3.2 Translating Structures into Guidance

The DMT-selected structures are translated into novice guidance using human-computer interaction (HCI) principles such as visibility, immediate feedback, and recognition [31]. In this work, the selected structures are implemented as three scaffolding layers, as shown in Fig. 2.

g_1 **Spatial corridor guidance:** The corridor encourages the end effector to follow a shared tube-like path structure. Motions inside the target corridor are visualised in green, while those outside are visualised in red.

g_2 **Gripper contact guidance:** This guidance encourages users to discover a robot-learnable grasp family and express it consistently. Two purple contact layers indicate the target grasp region, while green and red gripper visualisations indicate consistent and inconsistent contact orientations, respectively.

g_3 **Speed bar guidance:** This guidance provides pacing cues around task phases such as approach, descent, and contact. A speed bar gives immediate feedback: consistent phase-wise motion is visualised in green, while inconsistent pacing is visualised in red.

The numerical parameters used to realise these targets, such as corridor width, grasp-orientation tolerance, and speed-bar thresholds, are reported in Appendix A.5.

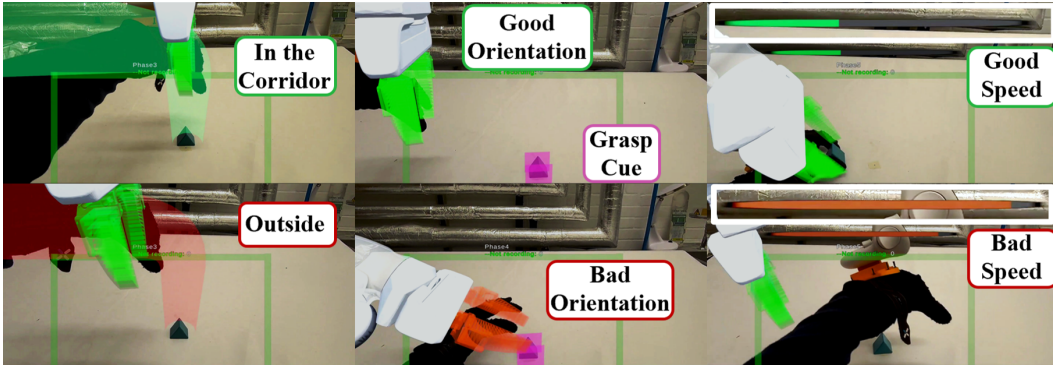


Figure 2: DMT-derived guidance in augmented reality (AR). Rows show guidance for good demonstrations (green) versus bad ones (red). The three columns present scaffold guidance in g_1 - g_3 . Each layer is shown in isolation without the previous guidance, keeping the intervention clean. Green boxes mark a shared recording region for clear point-cloud capture, not an additional DMT guidance.

4 Experiments

The following experiments evaluate whether DMT can train novice users to provide demonstrations that are more learnable for a fixed visuomotor policy learner under a finite demonstration budget. Specifically, three hypotheses are tested:

h_1 **Human success is not sufficient for robot learnability.** Task-successful human demonstrations are not necessarily robot-learnable under a fixed visuomotor learner and finite demonstration budget.

h_2 **Learner-aware teaching behaviour can be trained.** DMT-derived training targets can be translated into human-executable guidance that trains novice users to produce more robot-learnable demonstrations after guidance is removed.

h_3 **Learner-aware teaching behaviour can transfer.** The acquired teaching behaviour persists after guidance removal and transfers to unseen objects and related visuomotor manipulation tasks.

All comparisons within each experiment keep the learner, embodiment, action space, sensing pipeline, and budget fixed. Simulation and real-world experiments differ in sensing and collection interface. The setup is summarised in Appendix A.

4.1 Task Success and Robot Learnability

The first experiment tests h_1 by applying the DMT selection criterion from (4) in simulation. The goal is to evaluate whether different task-successful structures within $\mathcal{Z} = \mathcal{P} \times \mathcal{R} \times \mathcal{V}$ lead to different deployment performance under a fixed learner and finite demonstration budget. The simulation evaluates a candidate set $\mathcal{C} = \{z_0, z_{\mathcal{P}}, z_{\mathcal{R}}, z_{\mathcal{V}}\}$, where z_0 keeps the spatial, contact, and temporal structures consistent, while $z_{\mathcal{P}}$, $z_{\mathcal{R}}$, and $z_{\mathcal{V}}$ isolate variation in spatial path, contact orientation, and temporal pacing, respectively. The best evaluated candidate is defined as

$$\hat{z} = \arg \max_{z \in \mathcal{C}} \hat{J}(\mathcal{A}(\mathcal{D}_z)) \quad \text{s.t.} \quad |\mathcal{D}_z| = N. \quad (5)$$

Here, $\hat{z}$ denotes the best deployment-evaluated candidate within this finite structure set, and is used to identify the human training targets: spatial, contact, and temporal consistency.

Two skills are evaluated in PyBullet: rigid parallel-jaw grasping and peg insertion. For each task and condition, $N = 30$ task-successful scripted demonstrations are collected at 30 Hz.¹ Grasping and inserting are atomic skills in long-horizon manipulation tasks [32, 33, 34, 35]. Therefore, if task success is insufficient even for these atomic skills, it is unlikely to be sufficient as a general criterion for demonstration quality.

Four task-successful demonstration configurations are compared. The baseline z_0 keeps the spatial, contact, and temporal structures in s_1 – s_3 consistent. The other three variants change one structure at a time: approach geometry, gripper/contact orientation, or phase-wise pacing, while keeping the remaining structures fixed. This isolates spatial, contact, and temporal inconsistency as potential sources of learner incompatibility. Together, these comparisons test whether task completion alone is sufficient for robot learnability under a fixed learner, budget, and deployment protocol. The detailed configuration of each variant is provided in Appendix A.2.

4.2 Human Training through DMT Guidance

The next experiment tests h_2 and h_3 : whether guidance from DMT can train novices to provide more learnable demonstrations, and whether the acquired teaching behaviour transfers after guidance is removed. A between-subjects study is used with a target group and a control group. Both groups use the same AR interface, robot setup, tasks, and demonstration budget. Only the target group receives DMT-derived guidance during training; the control group receives matched practice without guidance.

Guidance. The guidance g_1 – g_3 are implemented as three scaffolding layers in §3.2 after evaluation from §4.1. Each guidance layer is practised for 30 demonstrations. These guided demonstrations are only used for human training and are not used for robot learning.

Tasks. The human training uses three tasks²: *grasping a prism*, *grasping a cube* and *ordered unseen-object pick-and-place*. Prism is used as the training task because it requires a consistent grasp family. The cube is used to test generalisation: although its geometry is similar, it still requires the same teaching principle of consistently demonstrating a robot-learnable grasp family. The longer-horizon ordered pick-and-place task is used as a broader post-curriculum generalisation test, where participants sequentially grasp a parallelepiped and a pentagram-shaped object and place them into a bowl. All tasks are visualised in Fig. 7.

Protocol. Each participant provides $N = 30$ unguided demonstrations for each evaluation phase, and one policy is trained per participant, task, and phase. From a power analysis (HHU G*Power) for detecting the effect size with a 0.1 error probability and 0.8 power, the study uses $n_p = 20$ novice participants, split equally between target and control groups³. The curriculum is summarised

¹The budget $N = 30$ was empirically selected as the smallest budget that reliably induced a high deployment performance.

²For all tasks, users begin from arbitrary positions within the workspace on their left-hand side, performing the task from left to right.

³This study was approved by the [Anonymised] (Approval No. [Anonymised]). Informed consent was obtained from all participants. All participants are novices with no prior background working with robots.

in Table 1. Hypothesis h_2 (improvement) is tested by comparing teaching improvements from P1 to P6 between target and control groups. h_3 (transfer) is tested by comparing improvements from P2 to P7 between target and control groups. P8 evaluates whether the teaching behaviour acquired on an atomic grasping skill generalises to longer-horizon manipulation tasks. Each participant-level policy is evaluated over $K = 10$ real-robot deployment rollouts. Robot starts are randomly sampled within the task workspace, and a rollout is counted as successful only if the task is completed without manual intervention, unsafe contact, object loss, or workspace-limit violation.

Phase	Task	Condition	Purpose
P1	Prism	No guidance	Trained-task pre-test
P2	Cube	No guidance	Unseen-object pre-test
P3	Prism	Spatial guidance / practice	Spatial training
P4	Prism	Gripper guidance / practice	Contact training
P5	Prism	Speed guidance / practice	Temporal training
P6	Prism	No guidance	Trained-task post-test
P7	Cube	No guidance	Unseen-object post-test
P8	Ordered pick-and-place	No guidance	Unseen-task test

Table 1: Evaluation protocol. The target group receives DMT-derived guidance in P3–P5, while the control group follows the same sequence of phases but without guidance. Only unguided phases (P1, P2, P6, P7 and P8) are used for robot learning.

4.3 Generic AR Feedback Baseline

The final experiment compares DMT-derived AR feedback with a generic AR feedback baseline. The goal is to test whether learner-aware feedback produces more learnable demonstrations than standard execution-oriented AR feedback. The same robot setup, teleoperation interface, tasks, learner, demonstration budget and practice as in §4.2 are used, with $n_p = 10$ new participants. Specifically, the baseline **ARCap-no-glove** uses demonstrations directly collected using generic AR feedback for execution support, including visibility, speed, and kinematic-feasibility cues. These demonstrations are compared with the post-training demonstrations from the DMT-target group (see Appendix A.7).

4.4 Evaluation Protocol and Metrics

The primary metric throughout the experiments is the empirical deployment success rate (EDSR), which empirically estimates the expected deployment objective J in (3). For a trained algorithm $\mathcal{A}(\mathcal{D}_z)$ from each participant

$$\hat{J}(\mathcal{A}(\mathcal{D}_z)) = \frac{1}{K} \sum_{k=1}^K \mathbf{S}(k), \quad (6)$$

where $\mathbf{S}(\tau_k) \in \{0, 1\}$ denotes k -th rollout success or fail under the deployment protocol. Tests use participant-level EDSR, averaged over $K = 10$ rollouts. Demonstration- and learning-level metrics are reported as secondary diagnostics to interpret how demonstration structure affects learner behaviour. These metrics include mean pointwise Cartesian variance (MPCV), mean closing orientation dispersion (MCOD), mean path-progress synchronisation variance (MPSV) and validation loss $\mathcal{L}$ for demonstration structures s_1 - s_3 , and diffusion denoising validation loss. Metrics and the corresponding detailed results are provided in Appendix B and C.1.

5 Results

5.1 Successful Demonstrations are Insufficient for Learnability (h_1)

Table 2 reports deployment success for policies trained from task-successful simulation demonstrations under the same learner and demonstration budget. For both grasping and inserting, the baseline achieves the highest EDSR, while the geometric, orientation, and temporal variants substantially reduce deployment performance. Against the baseline, Welch’s two-sided tests show significant drops in EDSR for all three variants in both tasks ($p < 0.001$). These results support h_1 : human task success is necessary but not sufficient for robot-learnable demonstration data under a fixed learner and

Condition	Grasping EDSR $\hat{J}$ (%)	Inserting EDSR $\hat{J}$ (%)
Baseline	87.5 ± 9.6	62.5 ± 9.6
Geometric variant	10.0 ± 8.2	12.5 ± 5.0
Orientation variant	2.5 ± 5.0	2.5 ± 5.0
Temporal variant	2.5 ± 5.0	0.0 ± 0.0

Table 2: Simulation evaluation of the $\mathcal{P}\text{-}\mathcal{R}\text{-}\mathcal{V}$ teaching-structure candidates under the fixed learner and budget. Each condition uses four independently scripted task-successful datasets, generated with different random seeds; each dataset contains $N = 30$ demonstrations and trains one policy from scratch. Values are reported as mean $\pm$ std across the four trained policies. EDSR is measured over $K = 10$ simulated deployment trials per dataset per task.

finite budget. Within the evaluated structure set, the $\hat{z}$ configuration yields the highest deployment performance, providing the deployment-evaluated target used to derive the training guidance.

5.2 DMT-trained Novices Produce More Learnable Data (h_2, h_3)

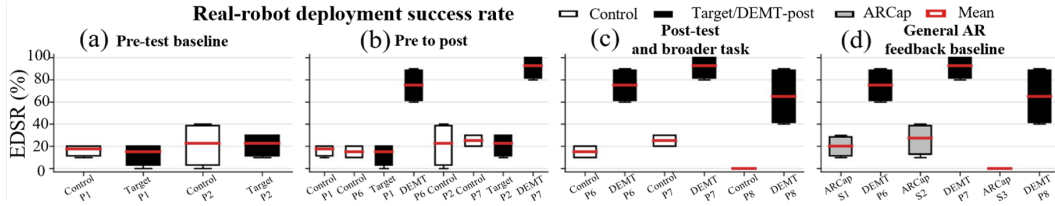


Figure 3: Real-robot EDSR. Each box summarises participant-level policy performance per group ($n = 10, K = 10$). (a) and (b) compare pre- and post-training performance on the trained prism task and the unseen cube task. (c) reports broader transfer to ordered pick-and-place. (d) compares DMT post-training demonstrations with a generic AR feedback baseline. White boxes denote control, black boxes denote target, grey boxes denote ARCap, and red horizontal bars denote means.

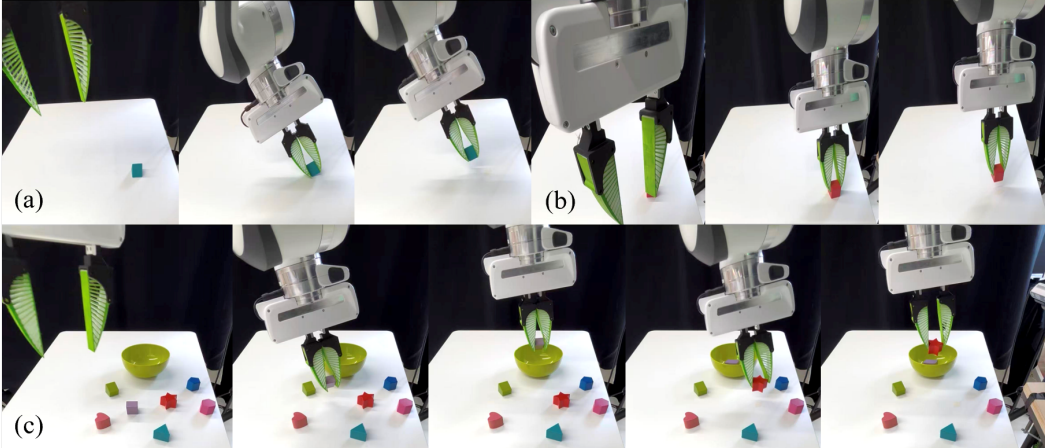


Figure 4: Qualitative real-robot deployment examples from policies trained on demonstrations provided by DMT-trained participants. Frames in each example are ordered from left to right. (a) Trained prism grasp-and-lift. (b) Unseen cube grasp-and-lift. (c) Ordered pick-and-place from clutter into the target bowl. The examples show that the deployment-success improvements correspond to successful closed-loop real-robot behaviours.

Fig. 3 reports robot deployment performance for policies trained from the unguided demonstrations in Table 1, where tests are two-sided, and results are treated as statistically significant when $p < 0.05$. Before training, the target and control groups show no significant difference on either the trained prism task (P1: $p = 0.708$) or the unseen cube task (P2: $p = 0.844$), confirming comparable pre-test performance. Since only unguided phases are used for policy training, any improvement in

P6 and P7 reflects changes in the participants’ teaching behaviour after guidance removal. Note that detailed demonstration- and learner-level results are reported in Appendix C.1.

Effectiveness of DEMENT-based training (h_2): For the trained prism task, the target group improves from $14.0 \pm 9.7\%$ deployment success in P1 to $75.0 \pm 10.8\%$ in P6. By contrast, the control group changes only from $17.0 \pm 4.8\%$ to $15.0 \pm 5.3\%$ after matched unguided practice. The improvement differs significantly between groups (two-sided Mann–Whitney, $U = 100$, $p < 0.001$). These results support h_2 , showing that DEMENT-derived guidance trains novices to provide more robot-learnable demonstrations after the guidance is removed.

Transfer of teaching behaviour (h_3): For the unseen cube task, the target group improves from $22.0 \pm 7.9\%$ deployment success in P2 to $91.0 \pm 8.8\%$ in P7, whereas the control group changes only from $22.0 \pm 14.8\%$ to $25.0 \pm 5.3\%$. The improvement differs significantly between groups (two-sided Mann–Whitney, $U = 100$, $p < 0.001$).

The ordered pick-and-place task further evaluates post-curriculum generalisation beyond atomic grasping. In P8, policies trained from DEMENT-trained participants achieve $55.0 \pm 12.7\%$ deployment success, while the control group were unable to teach the task with any success. The difference is significant (two-sided Mann–Whitney, $U = 100$, $p < 0.001$). Because these tasks are not used during the guided training phases, the results support h_3 , suggesting that the teaching behaviour acquired through DEMENT guidance transfers to an unseen object and to longer-horizon visuomotor manipulation.

5.3 Generic AR Feedback Baseline

The final comparison tests whether the improvement comes from DEMENT-derived demonstrator training rather than generic AR feedback alone. The **ARCap-no-glove** baseline achieves $20.0 \pm 8.2\%$, $27.0 \pm 12.5\%$, and $0.0 \pm 0.0\%$ deployment success on the prism, cube, and ordered pick-and-place tasks, respectively. By comparison, the **DEMENT-trained** group achieves $75.0 \pm 10.8\%$, $91.0 \pm 8.8\%$, and $55.0 \pm 12.7\%$ on the same tasks.

The DEMENT-trained group significantly outperforms the generic AR baseline on the trained prism task ($U = 100$, $p < 0.001$), the unseen cube task ($U = 100$, $p < 0.001$), and the ordered pick-and-place task ($U = 100$, $p < 0.001$). These results support that DEMENT trains human teaching better. Unlike ARCap-style task-level feedback, DEMENT-derived guidance trains novices to produce learner-relevant spatial, contact-orientation, and temporal structures, leading to more robot-learnable demonstrations after guidance is removed.

6 Conclusion

This paper introduced DEMENT, a deployment-evaluated framework for training novices to provide more robot-learnable demonstrations for visuomotor imitation learning. Rather than treating human task success as demonstration quality, DEMENT evaluates data through robot deployment under a fixed learner and finite budget. Experiments show that task-successful demonstrations can differ substantially in learnability because of spatial, contact-orientation, and temporal structure. DEMENT-derived guidance trains novices to preserve structural consistency after guidance is removed, improving real-robot deployment on the trained task, an unseen object, and a longer-horizon ordered pick-and-place task, and outperforming generic AR feedback.

7 Limitations and Future Work

This work has two main limitations. First, DEMENT is evaluated on single-arm, parallel-jaw, rigid-object manipulation tasks. The concrete spatial, contact-orientation, and temporal guidance designs are therefore specific to this setting. Future work should extend DEMENT to broader robot embodiments and task families, including dual-arm manipulation and potentially locomotion. Second, the current guidance is implemented through visual AR scaffolds. Other modalities, such as audio, haptics, text, or multimodal feedback, may provide more effective guidance for different users or tasks. Future work should compare alternative guidance designs across larger user populations.

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A Experimental Setup

A.1 General Experimental Setup

The experiments use a fixed Franka-based visuomotor learning pipeline across the simulation and human-training studies. The setup is summarised in Table 3.

Component	Exp A (Simulation)	Exp B (Human Training)
Platform	Franka Panda (PyBullet)	Franka Panda (Digital Twin)
Workspace	Simulated environment	Matched physical setup
Sensing	Virtual RGB-D	Intel RealSense D435
Calibration	Base-relative (Pre-set)	Base-relative (Calibrated)
Interface	Scripted generation	Meta Quest 3 AR (Streaming)
Back-end	PyBullet system	PyBullet + ARCap [7]
Observation	Virtual Point Cloud + Joint/Gripper states	Real Point Cloud + Joint/Gripper states
Action / Learner	Joint-angle chunks / 3D Diffusion Policy (Identical)	
Pipeline	Standard <i>robomimic</i> -based optimisation (Identical)	

Table 3: Common experimental setup. Identical components are merged to highlight the consistency between simulation and real-world implementations.

A.2 Simulation Experiment Variant Construction

The simulation used controlled task-success variants to test whether different s_1 - s_3 lead to different levels of robot learnability. As summarised in Table 4 and Fig. 5, each variant changed one structure while keeping the remaining structures fixed.

A.3 Data Collection System

The data collection system is shown in Fig. 6. The seat belt, adjustable chair, and shared recording box were used only to standardise the participant viewpoint and keep the target workspace within the RGB-D capture region across all groups. These controls were inherited from the AR data-collection setup and were not part of the DMT intervention. All participants were right-handed. Left-handed participants were excluded to avoid handedness-related variation in left-hand teleoperation proficiency. All participants therefore followed the same left-hand teleoperation protocol across groups.

Condition	Implementation	Isolated source
Baseline	Small approach region $[0.3, 0.5] \times [0.3, 0.5] \times [0.2, 0.5]$, fixed closing orientation $R_{\text{default}} = [-\pi, 0, 0]$, and fixed phase-wise pacing.	Reference
Geometric variant	Approach region enlarged to $[0.0, 0.5] \times [0.0, 0.5] \times [0.2, 0.7]$, while closing orientation and phase-wise pacing are kept fixed.	Spatial geometry
Orientation variant	Closing orientation alternates between R_{default} and $R_{\text{tilt}} = R_{\text{xyz}}(0^\circ, -45^\circ, 90^\circ)R_{\text{default}}$, while approach geometry and pacing are kept fixed.	Contact orientation
Temporal variant	Phase-wise temporal resampling varies the number of samples assigned to each phase, while preserving the spatial path and closing orientation.	Temporal pacing

Table 4: Task-success variants used in the simulation experiment. Each variant changes one structure while keeping the remaining structures fixed.

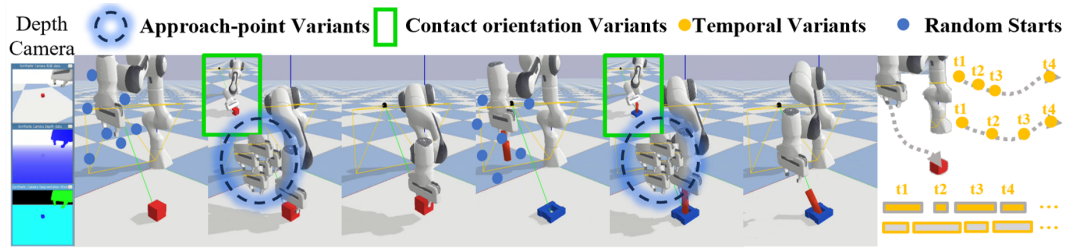


Figure 5: Simulation variant construction. The depth-camera panel shows RGB-D images used to construct the point cloud. Blue dots indicate randomly sampled feasible start states. The blue dashed circle shows the enlarged approach-point distribution for the geometric variant, and the green highlights show the contact-orientation variants. Yellow dots illustrate phase-wise temporal resampling while spatial path and contact orientation are held fixed.

391 A.4 Task Setup

392 Tasks 1 and 2 require participants to start from arbitrary positions within the left-hand side of the
393 shared human–robot workspace, grasp the target object, and lift it, using a prism and a cube, re-
394 spectively. The shared workspace denotes the region that is both comfortably reachable by the
395 participant’s left arm and reachable by the Franka Panda under the experimental table and safety
396 constraints. In the setup, the Franka base was positioned approximately 0.5 m from the participant
397 along the x direction. The Panda arm has a manufacturer-specified maximum reach of 855 mm.

398 A.5 Guidance Setup

Guidance	Parameter	Value
Spatial corridor	Radius	$r_{\text{start}} = 25\text{cm}, r_{\text{middle}} = 5\text{cm}, r_{\text{final}} = 3\text{cm}$
Grasp orientation	tolerance	15 deg
Speed bar	relative speed tolerance	$[0.5, 2.0] \times v_{\text{ref}}$

Table 5: Implementation parameters for the DMT-derived AR guidance. Here, the speed cue is normalised per task using the initial reference speed, rather than a fixed metric speed threshold: $v_{\text{ref}} = \|\mathbf{x}_1 - \mathbf{x}_0\|/(t_1 - t_0)$

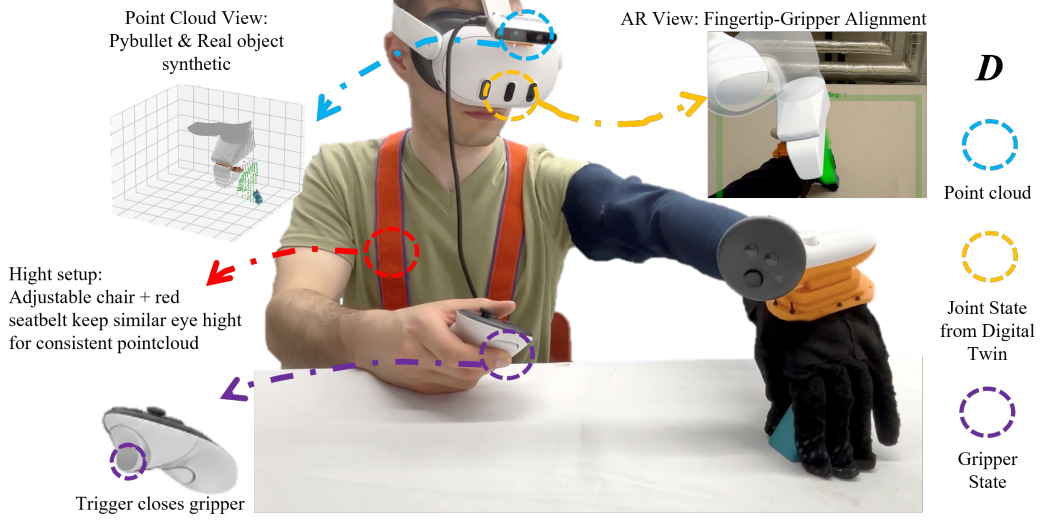


Figure 6: A participant is demonstrating; Demonstrations contain Point cloud converted from RGB-D frames (Blue), joint angles solved by the Pybullet back-end from the digital-twin end-effector pose (Yellow) and binary gripper state from the trigger (Purple). To keep a consistent camera view-point across participants from the same height (50 cm from desk to camera), the red seat belt and adjustable chair were used. A dark sleeve and glove were used for each participant to crop the real human arm in point cloud.

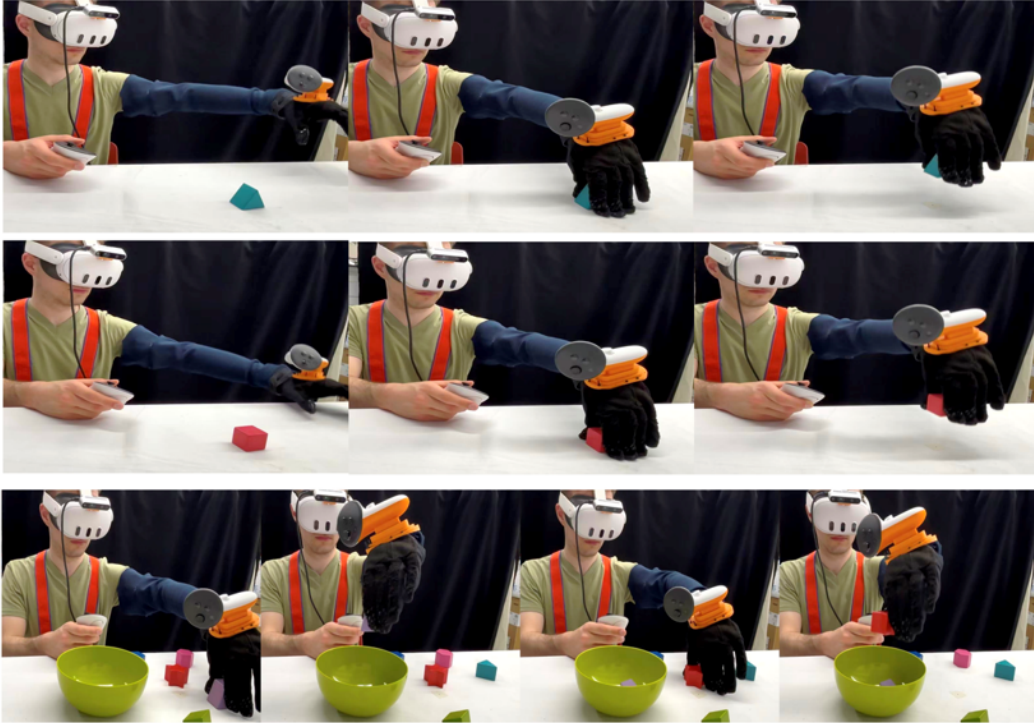


Figure 7: Human demonstration tasks used in the real-robot study. The first row shows the prism grasp task used for pre- and post-training evaluation on the trained object (P1/P6). The second row shows the cube grasp task used to evaluate transfer to an unseen object (P2/P7). The third row shows the longer-horizon ordered pick-and-place task (P8), where participants sequentially grasp unseen objects and place them into the target bowl. All demonstrations are collected through the same AR teleoperation interface, with the participant controlling the robot end-effector in AR and gripper state during task execution.

A.6 Human Training Evaluation Protocol

Table 6 summarises the human-training curriculum and the target-control comparison protocol. All human demonstrations were replayed and verified as task-successful demonstrations before robot learning.

Group	n_p	P1	P2	P3	P4	P5	P6	P7	P8
Target	10	Prism No guide Learning	Cube No guide Learning	Prism Spatial Practice	Prism Contact Practice	Prism Temporal Practice	Prism No guide Learning	Cube No guide Learning	Ordered No guide Learning
Control	10	Prism No guide Learning	Cube No guide Learning	Prism No guide Practice	Prism No guide Practice	Prism No guide Practice	Prism No guide Learning	Cube No guide Learning	Ordered No guide Learning

Table 6: Human-training protocol for the target and control groups. Both groups contain $n = 10$ novice participants and use the same tasks, AR interface, robot setup, and demonstration budget. Each phase contains $N = 30$ demonstrations. The target group receives DEMENT-derived spatial, grasp, and pacing guidance in P3–P5, respectively, while the control group performs matched unguided practice. Only unguided evaluation phases, P1, P2, P6, P7, and P8, are used for robot learning. Guided or practice demonstrations in P3–P5 are used only for human training.

A.7 AR Guidance Comparison

Participants in the ARCap baseline were given equivalent pre-collection practice for a fair comparison with DEMENT. Table 7 and 8 summarise the visible guidance used in the generic AR feedback baseline and in DEMENT-derived training. The ARCap baseline feedback is given mainly through colour-frame warnings for view, speed, and collision during collection [7].

Baseline	n_p	Prism	Cube	Ordered pick-place
ARCap-no-glove	10	Generic AR feedback 30 demos Policy data	Generic AR feedback 30 demos Policy data	Generic AR feedback 30 demos Policy data

Table 7: Generic AR feedback baseline used for comparison with DEMENT-trained post-training demonstrations. The ARCap-no-glove baseline uses the same participant budget and policy-training budget, but provides execution-oriented AR feedback during data collection rather than pre-collection demonstrator training.

	ARCap-no-glove	DEMENT-guidance
Visual guidance	Colour frame only	Corridor + grasp + pacing
Meaning	View / speed / collision	Spatial / contact / temporal consistency
When	During collection	Before collection
Goal	Improve task execution	Improve novice teaching

Table 8: Difference between the generic AR feedback baseline and DEMENT-derived guidance. ARCap-no-glove uses colour-frame warnings to support task execution during collection, whereas DEMENT uses corridor, grasp, and pacing guidance to train novices before collection.

A.8 3D Diffusion Policy Setup

We use a 3D Diffusion Policy with a PointNet encoder for point-cloud observations and a 1D conditional U-Net for joint-space action prediction. All experiments use the same learner and hyperparameters: 2000 sampled XYZ-RGB points, $T_o = 5$, $T_p = 20$, batch size 16, 3000 epochs, random seed as 1, Adam with learning rate 10^{-4} , DDIM with 100 training diffusion steps and 10 inference steps, and loss $0.7\mathcal{L}_{arm} + 0.3\mathcal{L}_{hand}$.

A.9 Rollout Setup

Each trained policy is evaluated with 10 rollouts from random initial end-effector states sampled from $[0.0, 0.6] \times [-0.1, 0.1] \times [0.0, 0.6]$ m relative to the robot base. During rollout, the controller runs at 30 Hz; the policy uses the latest 5 observation frames to predict 1 arm-gripper action, which is held for 5 control steps before replanning.

B Evaluation Metrics

The main paper reports empirical deployment success rate as the primary metric. Additional demonstration-level and learner-level diagnostics are used to analyse the evidence chain from demonstration structure to learner behaviour and deployment performance.

B.1 Demonstration-Level Structure Metrics

All trajectories are temporally normalised to $T = 100$ steps over normalised time in $[0, 1]$. The following metrics quantify the three human-controllable structures used in DEMENT: spatial path consistency, contact-orientation consistency, and temporal consistency.

Mean Path Coordinate Variance. The MPCV measures spatial path consistency:

$$\text{MPCV} = \frac{1}{T} \sum_{t=1}^T \text{Var}_i \left(\mathbf{p}_t^{(i)} \right), \quad (7)$$

where $\mathbf{p}_t^{(i)}$ denotes the end-effector position of demonstration i at normalised time t . Lower MPCV indicates more consistent path geometry across demonstrations. The unit is m^2 .

Mean Closing Orientation Dispersion. The MCOD measures contact-orientation consistency around the gripper-closing event:

$$\text{MCOD} = \frac{1}{|\mathcal{W}|} \sum_{\tau \in \mathcal{W}} \left\langle d_q \left(q_{c_i+\tau}^{(i)}, q_{c_j+\tau}^{(j)} \right) \right\rangle_{i < j}, \quad d_q(q_a, q_b) = 2 \arccos(|q_a^\top q_b|). \quad (8)$$

Here, $q_{c_i+\tau}^{(i)}$ denotes the end-effector orientation quaternion of demonstration i at offset τ around its gripper-closing index c_i , and $\mathcal{W}$ is a fixed temporal window around gripper closing. Lower MCOD indicates more consistent contact-orientation strategy. The unit is radians.

Mean Path-progress Synchronisation Variance The MPSV measures temporal consistency:

$$\text{MPSV} = \frac{1}{T} \sum_{t=1}^T \text{Var}_i(\rho_t^{(i)}), \quad \rho_t^{(i)} = \frac{\sum_{k=1}^{t-1} \|\mathbf{p}_{k+1}^{(i)} - \mathbf{p}_k^{(i)}\|_2}{\sum_{k=1}^{T-1} \|\mathbf{p}_{k+1}^{(i)} - \mathbf{p}_k^{(i)}\|_2}. \quad (9)$$

Here, $\rho_t^{(i)}$ denotes the fraction of the end-effector path completed by demonstration i at normalised time t . Lower MPSV indicates more consistent speed and phase timing across demonstrations. This metric has no unit because it is a ratio.

B.2 Learner-Level Diagnostic Metric

Validation loss is reported as a learner-level diagnostic using the fixed 3D Diffusion Policy objective:

$$\mathcal{L} = \lambda_{\text{arm}} \|\hat{\epsilon}_{\theta}^{\text{arm}} - \epsilon^{\text{arm}}\|_2^2 + \lambda_{\text{grip}} \|\hat{\epsilon}_{\theta}^{\text{grip}} - \epsilon^{\text{grip}}\|_2^2, \quad \lambda_{\text{arm}} = 0.7, \lambda_{\text{grip}} = 0.3. \quad (10)$$

At each training step, Gaussian noise is added to expert action chunks at sampled diffusion timesteps, and the network is trained to predict the injected noise. The loss is computed as a weighted mean-squared error between predicted and true noise, with separate terms for arm joint angles and gripper states. Validation loss is reported on held-out demonstrations to measure how well the fixed learner fits the demonstrated action distribution. It is treated as a secondary metric because low validation loss does not necessarily imply high deployment success.

C Experiments Results

C.1 Extra Quantitative Analysis

Table 9 reports detailed demonstration-level and learner-level results. The DEMENT post-training and transfer conditions achieve the lowest structural variability and validation loss across all tasks.

Phases	MPCV(m^2)	MCOD(rad)	MPSV	Val $\mathcal{L}$
Target P1 prism pre	0.0046 ± 0.0016	0.3342 ± 0.0709	0.0026 ± 0.0004	0.2646 ± 0.1240
DEMT P6 prism post	0.0007 ± 0.0004	0.0749 ± 0.0051	0.0013 ± 0.0003	0.0710 ± 0.0230
Control P1 prism pre	0.0052 ± 0.0009	0.3732 ± 0.0382	0.0028 ± 0.0004	0.2316 ± 0.0653
Control P6 prism post	0.0053 ± 0.0007	0.3757 ± 0.0228	0.0031 ± 0.0002	0.2252 ± 0.0886
ARCap prism	0.0055 ± 0.0002	0.3704 ± 0.0156	0.0027 ± 0.0004	0.1896 ± 0.0638
Target P2 cube pre	0.0083 ± 0.0014	0.6582 ± 0.1383	0.0056 ± 0.0017	0.2460 ± 0.0650
DEMT P7 cube post	0.0005 ± 0.0003	0.1668 ± 0.0296	0.0010 ± 0.0002	0.0606 ± 0.0084
Control P2 cube pre	0.0086 ± 0.0008	0.6806 ± 0.0632	0.0058 ± 0.0016	0.2570 ± 0.0733
Control P7 cube post	0.0088 ± 0.0011	0.6809 ± 0.0623	0.0055 ± 0.0028	0.2295 ± 0.0676
ARCap cube	0.0070 ± 0.0007	0.5727 ± 0.0755	0.0053 ± 0.0017	0.2153 ± 0.0753
DEMT P8 ordered	0.0008 ± 0.0005	0.0698 ± 0.0036	0.0005 ± 0.0002	0.0395 ± 0.0203
Control P8 ordered	0.0052 ± 0.0011	0.2762 ± 0.0511	0.0017 ± 0.0005	0.2726 ± 0.0458
ARCap ordered	0.0044 ± 0.0007	0.2503 ± 0.0699	0.0015 ± 0.0007	0.2973 ± 0.0617

Table 9: Demonstration- and learner-level results for the real-robot study. Rows report mean $\pm$ standard deviation for spatial path consistency (MPCV), contact-orientation consistency (MCOD), temporal synchronisation (MPSV), and validation loss of the fixed 3D Diffusion Policy. Lower values indicate more consistent demonstrations or better learner fit. Across the trained prism task, unseen cube task, and ordered pick-and-place transfer task, DEMENT-trained post-training demonstrations show lower structural variability and lower validation loss than the matched control and generic ARCap baselines. Bold entries highlight the DEMENT post-training or transfer conditions.

C.2 Failure Analysis

The failure analysis qualitatively explains the participant-level EDSR results reported in Fig. 3. Qualitative rollout analysis in Fig. 8 shows that most failures were structured rather than random.

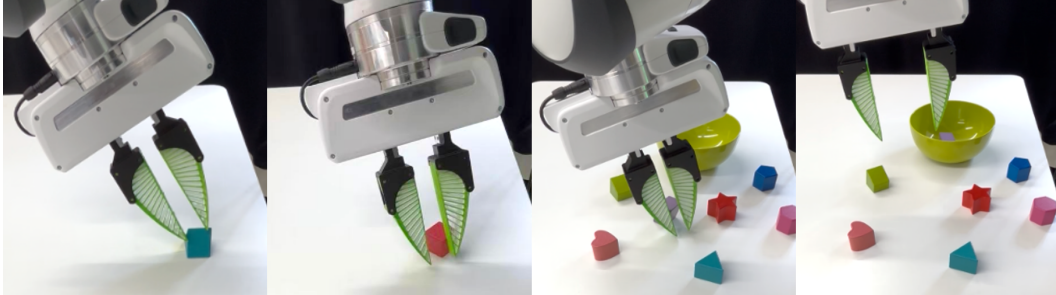


Figure 8: Real-robot DEMENT-failure cases. Most failures occur at contact-critical grasp transitions, while rare long-horizon failures occur at phase transitions between phases.

In the grasping tasks, the failed policies often touched the object but tilted or displaced it instead of forming a stable grasp. For the cube, failures often involved off-centre contact, causing edge slip during closure or early lift. These failures suggest that the policy learned coarse reaching, but remained sensitive to contact factors such as final approach alignment, grasp orientation, and closure–lift timing.

The ordered pick-and-place task showed a similar bottleneck. Most failures selected and reached the correct target, but failed to convert contact into a stable grasp. A few failures were partial successes with sequencing errors. In one rare case, the policy completed the purple phase but then made small hesitation motions near the bowl without committing to the next target. Overall, the common failures occurred at contact-critical grasp transitions, while rare long-horizon failures occurred at phase transitions.